MONOMORPHISMS DESCEND TO A FINITE LEVEL VIA THE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. No omissions: the reduction to the diagonal is explicit and the permissibility citations are given.

Allowed standard prerequisites (editorial).

- Monomorphisms of schemes. Schemes, Lemma schemes-lemma-monomorphism
- Descend isomorphism. Limits, Lemma lemma-descend-isomorphism (same chapter)

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01L3: sources/stacks-chow/tags.txt, source lines 2072–2072. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0818: sources/stacks-chow/tags.txt, source lines 10208–10208. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0816: sources/stacks-chow/tags.txt, source lines 10206–10206. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081E: sources/stacks-chow/tags.txt, source lines 10214–10214. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Situation 081D. If*

(1) *f is a monomorphism, and*
(2) *f_0 is locally of finite type,*

then f_i is a monomorphism for some $i \geq 0$.

Proof. Recall that a morphism of schemes $V \to W$ is a monomorphism if and only if the diagonal $V \to V \times_W V$ is an isomorphism (Schemes, Lemma 01L3). The morphism $X_0 \to X_0 \times_{Y_0} X_0$ is locally of finite presentation by Morphisms, Lemma 0818. Since $X_0 \times_{Y_0} X_0$ is quasi-compact and quasi-separated (Schemes, Remark 0816) we conclude from Lemma 081E that $\Delta_i : X_i \to X_i \times_{Y_i} X_i$ is an isomorphism for some $i \geq 0$. For this i the morphism f_i is a monomorphism. □

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